

Bishoy Amgad

bishoy.amgad71@gmail.com — +20 1228898234 — Cairo, Egypt
linkedin.com/in/bishoy-amgad-5843902a2 — github.com/bishoy-512

Objective

3rd-year Computer and Information Sciences student at Ain Shams University with hands-on experience building Retrieval-Augmented Generation (RAG) pipelines using FastAPI, MongoDB, Qdrant, and LangChain, and experimenting with both cloud (Cohere) and local (Ollama) LLM providers. Seeking an AI Knowledge Systems & RAG Engineering internship to apply this experience to building enterprise document-intelligence and knowledge-base solutions with open-source AI tools.

Education

Ain Shams University, Faculty of Computer and Information Sciences

Cairo, EG

Bachelor's in Computer and Information Sciences - 3rd Year

2023 – 2027

Skills

Programming Languages: C, C++, Python, Java, SQL, Assembly

Data Science & ML: EDA, Feature Engineering, Regression, Classification, Clustering, Model Evaluation, NumPy, Pandas, scikit-learn, Matplotlib, Streamlit

Deep Learning & Tools: TensorFlow, Keras, CNNs, Data Augmentation, Jupyter Notebook/Colab

NLP & Text Processing: Text Preprocessing, Tokenization, Stemming/Lemmatization, TF-IDF, Word Embeddings (Word2Vec, GloVe), Sentiment Analysis, HuggingFace Transformers, NLTK, spaCy

Databases & Version Control: MySQL, SQL Queries, Git, GitHub, Linux (Basic), VS Code

Core Competencies & Competitive Programming: OOP, Data Structures, Algorithms, Problem Solving, Operating Systems, Computer Architecture, Solved 300+ problems on [Codeforces](https://codeforces.com/)

Projects

Mini-RAG: Retrieval-Augmented Generation System

Python, FastAPI, MongoDB, Qdrant, LangChain, Ollama

- Built a RAG pipeline end-to-end (FastAPI backend, MongoDB storage, Qdrant vector indexing) with document ingestion, chunking, and semantic search over PDFs, and switchable Cohere/Ollama embedding providers for cloud vs. local LLM use.

Loan Completion Status Prediction

Python, scikit-learn, Pandas, SMOTE

- Built a 4-class Random Forest classifier achieving 96.36% accuracy and 0.96 macro F1-score across all classes using SMOTE on a 1,810-record dataset.

Emotion Detection from Text

Python, NLTK, scikit-learn, NLP

- Built a multi-class text classifier to detect emotions (Happy, Fear, Sad) using NLP preprocessing techniques including tokenization, stemming, and TF-IDF vectorization, trained and evaluated on labeled text data.

ML Algorithms from Scratch

Python, NumPy

- Implemented 4 core machine learning algorithms (SVM, KNN, Linear and Logistic Regression) from scratch using NumPy, covering optimization, distance metrics, and gradient-based learning.

Flower Image Classification

Python, TensorFlow, Keras, OpenCV

- Engineered a CNN model for image recognition, utilizing Data Augmentation and OpenCV to enhance generalization on unseen data.

Library Management System

Java, OOP

- Designed a Java OOP-based library system managing 5+ core entities using Collections and custom exception handling for reliable operations.

Gym & Padel Management System

C++, OOP, Data Structures

- Designed and implemented a C++ OOP-based management system with efficient booking logic using Trees, Queues, and Linked Lists for fast search and scheduling.

Fundamentals of Operating Systems (FOS)

C

- Implemented core OS components in C, including dynamic memory allocation, kheap/uheap, shared memory with locks, CPU scheduling, and 5 page replacement algorithms (Placement, LRU, Optimal, Clock, MClock).

Certificates

AI & ML Foundations – Sprints — **Machine Learning** – ApplAi

- Covered AI/ML fundamentals, supervised/unsupervised learning, and model evaluation techniques.
- Focused on advanced ML techniques, feature engineering, and real-world ML applications.

Deep Learning – Nvidia — **Deep Learning** – Huawei

- Specialized training in neural networks, CNNs, and deep learning frameworks (Nvidia).
- Advanced deep learning concepts and practical implementation (Huawei).

Generative AI – ITI & NVIDIA

- Generative AI, prompt engineering, and LLM fundamentals.

Languages

Arabic: Native

English: Intermediate